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Automated detection and classification of leukemia on a subject-independent test dataset using deep transfer learning supported by Grad-CAM visualization

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ABSTRACT

Leukemia is a type of cancer that affects blood cells and causes fatal infection and premature death. Modern technology enabled by the machine and advanced deep learning algorithms can assist doctors in diagnosing the disease in the early stage. Automatic classification of disease based on advanced deep learning always require a huge number of labeled dataset. This paper presents an image dataset of 750 microscopic blood smears. The dataset contains images of Chronic Lymphocytic Leukemia, Acute Lymphoblastic Leukemia, Chronic Myeloid Leukemia, and Acute Myeloid Leukemia. The dataset is enlarged by merging images of a dataset present in the literature, containing 500 images of Acute Lymphoblastic Leukemia, Acute Myeloid Leukemia, and normal cases. This dataset is used for automated detection and classification of leukemia using deep transfer learning, which is the major task of the proposed work. The impact of subject-independent test dataset on the robustness of trained models is also analyzed on different data splits for training and testing. For the classification task, the proposed work obtained an accuracy of 84% on a subject-independent test dataset when support vector machine is trained on features extracted by a pruned VGG16 whose last three convolutional layers are fine tuned. The features responsible for classifying an image to a particular class are visualized with the help of Gradient-weighted Class Activation Mapping technique. The dataset is being created by taking the opinion of various experts that will assist the scientific community in conducting medical research supported by machine learning models.

1. Introduction

Leukemia is a type of cancer related to blood cells due to which they lose their ability to function normally. Abnormal blood cell count increases in large numbers compared to normal ones in the bone marrow, causing life-threatening infection and in some cases, premature death [1,2]. Leukocytes, commonly known as WBC, become abnormal both morphologically as well as functionally and the proliferation becomes aberrant. Leukemia can be detected in any age group but commonly younger people under 15 and adults over 55 get affected. Patients with leukemia may exhibit symptoms similar to common illnesses like flu, such as fever, fatigue, weakness, or aches in the bones and joints. This disease cannot be diagnosed and sub-typed easily as the abnormal WBC can sometimes be difficult to distinguish morphologically from normal WBC on microscopy. Leukemia is generally diagnosed by carrying out various tests like blood counts, peripheral smear examinations, lumbar puncture for cerebrospinal fluid study, bone marrow analysis, and myelograms [3]. These tests except for the blood counts are performed by pathologists manually and hence the reliability of the tests relies upon the experience and fatigue level of the pathologist. Therefore, there is a need to develop a computer-aided automatic system for accurate and precise detection and classification of leukemia. This will help in reducing the burden of doctors and pathologists, and aid detection of the disease at an early stage too.

The process of formation of blood cells inside bone marrow is called hemopoiesis. In their initial stage blood cells are called stem cells which get differentiated into WBC, platelets, and RBC. Stem cells are of two types, myeloid stem cells, and lymphoid stem cells, as shown in Fig. 1. Myeloid stem cells are responsible for the formation of RBC, some types of WBC (monocytes, eosinophils, neutrophils, and basophils), and platelets while lymphoid stem cells develop into other types of WBC,

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Abbreviation

WBC	White Blood Cell
CLL	Chronic Lymphocytic Leukemia
CML	Chronic Myeloid Leukemia
LBP	Local Binary Pattern
MLP	Multilayer Perceptron
KNN	K-Nearest Neighbors
NB	Naive Bayesian
CNN	Convolutional Neural Network
AC-GAN	Auxiliary Classifier Generative Adversarial Network
EDTA	Ethylene Diamine Tetraacetic Acid
TIFF	Tagged Image Format File
FCL	Fully Connected Layers
Grad-CAM	Gradient-weighted Class Activation Mapping
SC	Smudge Cell
GLCM	Gray Level Co-occurrence Matrix
RBC	Red Blood Cell
ALL	Acute Lymphoblastic Leukemia
AML	Acute Myeloid Leukemia
RF	Random Forest
SVM	Support Vector Machine
RBFN	Radial Basis Function Network
MM	Multiple Myeloma
IoMT	Internet of Medical Things
AIIMS	All India Institute of Medical Sciences
ROI	Region of Interest
EF	Extracted Feature
LTCL	Last Three Convolutional Layers
t-SNE	t-distributed Stochastic Neighbour Embedding
FAB	French–American–British
WHO	World Health Organisation

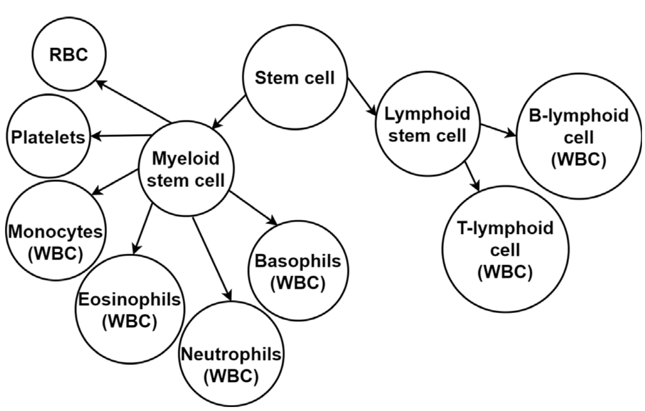


Fig. 1. Blood cells formation.

B-lymphoid cells and T-lymphoid cells [4]. When WBC originating from myeloid stem cell becomes cancerous then this type of leukemia is called myeloid leukemia and the one from lymphoid stem cell is called lymphocytic or lymphoblastic leukemia. Depending on the rate of growth of cancerous cells and the stage of differentiation, leukemia is again of two types. When cancerous cells increase rapidly with the increased number of blasts, then it is a case of acute leukemia and if growth is slow with more maturing cells and less than 20% blasts then it is a case of chronic leukemia. Therefore, leukemia can be categorized into four categories which are CLL, ALL, CML, and AML [5]. However, chronic leukemia not detected at an early stage and left untreated can transform into an acute one. In some cases, chronic leukemia can transform into acute one in a very short interval of time, even, within a week [6,7]. Although acute and chronic leukemias can occur in any age group, acute leukemia is generally more common in paediatrics and chronic leukemia mainly affects adults and old people [8]. Due to leukemia, above 3 lac people died globally and 32,471 death occurred in India, as per GLOBOCAN 2018 report [9].

With computer-aided systems, automatic detection and classification of leukemia can be done. This will help in reducing the chances of error in diagnosing the disease and the load on doctors.

1.1. Related work

Several methods have been proposed for the detection of leukemia and classification of its types, especially, ALL and AML. Table 1 presents studies related to detection of leukemia and highlights feature extraction techniques, classifier used, number and type of images considered in the studies, and accuracy obtained in the classification of leukemia. Mohapatra et al. [10,11], Singhal and Singh [12], Putzu et al. [13], Rawat et al. [14], and Neoh et al. [15] use images containing only one WBC to detect leukemia. Madhukar et al. [16], Patel and Mishra [17], Vincent et al. [18], Agaian et al. [19], and Dese et al. [20] use images containing more than one WBC to detect leukemia. These works detect leukemia by extracting features from the images and using a classifier to classify the input image. Combination of features like texture (Haralick's features [27] and LBP [28]), color features (uniformity, mean, entropy, and standard deviation), and shape (perimeter, Fractal Dimension [29], area, and compactness) are used to train the classifier like RF [30], MLP [31], SVM [32], KNN [33], RBFN [34] and NB [35] to detect leukemia. However, the performance of these methods mainly relies on feature selection which is done manually.

Recent works report that deep learning technique can do the feature extraction automatically [36,37]. Vogado et al. [2] present a method to detect leukemia using three different databases. They use transfer learning technique along with an SVM classifier for classification. They achieve an accuracy of more than 99%. Kumar et al. [21] represent a binary classification of MM and ALL using a CNN model. The dataset consists of 100 MM and 90 B-ALL images. An accuracy of 97.2% is achieved for the binary classification. Abhishek et al. [22] introduce a dataset of 500 microscopic blood images. The dataset includes three types of images, i.e., normal, AML, and ALL images. They enlarge the image dataset by merging images of ALL-IDB [38]. They perform binary and three-class classifications using the transfer learning technique. Accuracy of 98% for binary classification and 95% for three-class classification are achieved. Bukhari et al. [23] propose a deep learning method that incorporates squeeze and excitation-based learning technique to detect leukemia. They achieve an accuracy of 98.3% when ALL_IDB1 and ALL_IDB2 [38] are used to evaluate their methodology. Karar et al. [24] present an intelligent IoMT framework to do classification and detection of leukemia. They develop an AC-GAN classifier and achieve an accuracy of 98.67% for classifying an image into ALL or healthy category and 95.5% for classifying an image into ALL, AML, or normal class. Das et al. [25] apply transfer learning technique to detect and classify leukemia using three different datasets, ALLIDB1 [38] containing 49 microscopic images of ALL cases and 59 microscopic images of normal cases, ALLIDB2 [38] containing 130 microscopic images of ALL cases and 130 microscopic images of normal cases, and 40 microscopic images of non-AML cases and 40 microscopic images of AML cases collected from the ASH database [39]. They apply MobileNetV2 [40] for feature extraction and SVM as a classifier. They obtain 99.39%, 98.21%, and 97.73% accuracies for the classification

Table 1
Studies related to detection of leukemia.

Work	EF and Classifier	Image type	No. of images	Accuracy (%)
	Single WBC image			
Mohapatra et al. [10]	Hausdroff dimension, contour signature, shape features & color features along with SVM	ALL & normal	108	93.00
Mohapatra et al. [11]	Morphological features, Haar wavelet, Haralick's feature, 2-D discrete Fourier transform & color features along with ensemble of SVM, RBFN, MLP, NB & KNN	ALL & normal	104	94.73
Singhal and Singh [12]	LBP & GLCM along with SVM	ALL & normal	260	93.84
Putzu et al. [13]	Haralick's features, shape features & color features along with SVM	ALL & normal	267	93.63
Rawat et al. [14]	Shape features, signal processing based features, statistical methods, color features along with SVM	ALL, AML & normal	420	99.50
Neoh et al. [15]	Shape, color & texture features along with Dempster-Shafer ensemble	ALL & normal	180	96.72
	Multiple WBC image			
Madhukar et al. [16]	Hausdroff dimension, GLCM, shape & color features along with SVM	ALL & normal	98	93.50
Patel and Mishra [17]	Shape, color, statistical and texture features along with SVM	ALL & normal	27	93.57
Vincent et al. [18]	GLCM, Hausdorff Dimension & geometrical features along with MLP	ALL, AML & normal	100	97.70
Agaian et al. [19]	Haralick's features, Hausdorff dimension, Haar wavelet, shape & color features along with SVM	ALL & normal	98	> 94.00
Dese et al. [20]	Texture, geometric, statistical, and color features along with multi class SVM	AML, ALL, CML, CLL & normal	520	97.69
	Deep learning			
Vogado et al. [2]	Vgg-f along with SVM	ALL & normal	754	99.20
Kumar et al. [21]	CNN	MM & B-ALL	190	97.20
Abhishek et al. [22]	Pre-trained ResNet50 along with SVM	ALL, AML & normal	500	95.00
Bukhari et al. [23]	Squeeze and excitation-based CNN	ALL & normal	368	98.30
Karar et al. [24]	AC-GAN	ALL, AML & normal	445	95.50
Das et al. [25]	MobileNetV2 along with SVM	ALL & normal	108	99.39
		ALL & normal	260	98.21
		AML & normal	80	97.73
Baig et al. [26]	Concatenated features along with bagging ensemble	MM, AML & ALL	452	97.04

Table 2
Existing image datasets and proposed dataset used for detection of leukemia.

Authors	Image type	Image size	Size of dataset
Agaian et al. [41]	AML & normal	184 × 138	80
Labati et al. [38]	ALL & normal	2592 × 1944	108
Kumar et al. [21]	B-ALL & MM	2560 × 1920	190
Abhishek et al. [22]	ALL, AML & normal	4908 × 3264	500
Proposed dataset	**ALL, AML, CLL, CML & normal**	**4908 × 3264**	**1250**

tasks corresponding to ALLIDB1, ALLIDB2, and ASH databases, respectively. Baig et al. [26] employ transfer learning technique to detect MM, AML, and ALL. They construct an image dataset containing 4150 microscopic images. The dataset consists of 114 images belonging to MM, 31 images belonging to ALL, and 4013 images belonging to AML. To handle the problem of class imbalance 307 images of AML cases are randomly chosen and to handle the overfitting problem data augmentation is performed on MM and ALL images. After doing data augmentation the count of MM images increased to 301 and that of ALL increased to 293. After doing pre-processing of images, enriched segmented images are obtained. The segmented images are given to two CNNs for feature extraction and extracted features are concatenated using the canonical correlation analysis technique. The concatenated features are used for the training of different classifiers. Bagging ensemble achieves an accuracy of 97.04% for the classification task.

1.2. Motivation and contribution

From Table 2 and the literature, it can be observed that most of the work is limited to the classification of acute type of leukemia. Chronic leukemia cells grow slowly but if not detected and treated on time then it is observed that in some cases (less than 2%) chronic leukemia can progress into acute type [42–45]. From the literature, it can also be inferred that recent works based on deep learning technique perform better than traditional image processing and manual feature extraction methods to detect leukemia. However, most of the work does not mention whether the images considered for the evaluation of trained models are subject-independent or not. If the images considered in the test dataset belong to subjects whose images are also present in the training dataset then this may impact the robustness of the trained model. Another major problem with deep learning technique is the unavailability of huge and well-annotated datasets, especially in the biomedical domain [2,46,47]. The present work addresses the mentioned challenges. The contributions of this article are as follows.

1. A new microscopic blood image dataset of CLL, ALL, CML, and AML is proposed. The proposed dataset contains 750 images and the dataset is enlarged by combining the 500 images present in the dataset of Abhishek et al. [22].
2. The combined dataset now contains 1250 images and this dataset is used for the detection and classification of leukemia with the help of deep transfer learning method.
3. The impact of the subject-independent test dataset on the robustness of trained models is analyzed on two different data splits, *i.e.*, 80% training and 20% testing, and 50% training and 50% testing. A subject-independent test dataset contains images that do not belong to subjects whose images are present in the training dataset.
4. The classification results obtained in the study are analyzed qualitatively as well as quantitatively, and the features responsible for classifying an image to a particular class are visualized with the help of class-specific heatmaps.

The following is the layout of the paper. Section 2 introduces the proposed combined dataset. The Section 3 discusses deep transfer

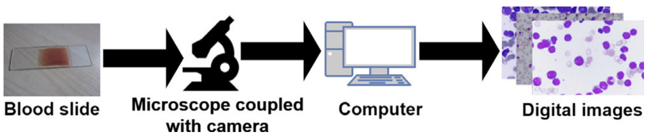


Fig. 2. Dataset preparation.

Table 3
Image counts belonging to specific magnification and category.

Magnification	Normal	CLL	ALL	CML	AML	Total
400×	26	28	23	21	23	121
1000×	224	222	227	229	227	1129
Total	250	250	250	250	250	1250
No. of patients	20	7	15	5	19	66

learning method for the detection of leukemia and its types. The results of the experiments are detailed in Section 4, and Section 5 presents the conclusion of the work.

2. Dataset

To detect and classify leukemia using deep transfer learning technique, a microscopic peripheral blood smear image dataset is formed. There are 1250 microscopic images of normal, CLL, ALL, CML, and AML cases which are collected from 66 subjects. Images were collected as part of a clinical test that ran between May 2019 and February 2022 in collaboration with AIIMS Patna, India. This study has been carried out with the approval of institutional ethics committees and following ethical norms. All participants and their caregivers gave their informed consent. Normal case samples were obtained in association with the Department of Physiology, AIIMS Patna. The Hematology section of the Department of Pathology, AIIMS Patna helped in obtaining a morphological diagnosis of cancerous samples. The location of the possible cancerous cells and the subcategory of leukemia were determined by considering the opinions of two experts. If both the experts agree with each other then only a sample was included in the study and the image was labeled as per the suggestions of the experts. If there was any difference in the opinions of the experts regarding a sample then it was not considered in this study. All the information was anonymous.

The pathologist uses the conventional approach to prepare the peripheral blood smear slides. In an EDTA vacutainer, a blood sample of 2 mL is collected. Smears are made by putting a blood drop on a clean slide and spreading it according to the wedge technique. After that, the smear is air dried. After three minutes, the smear is dyed with Leishman stain, and then by Giemsa stain (1 : 10 dilutions; 15 min). The stained slide is cleansed under flowing tap water and then left to dry [48].

Fig. 2 shows the dataset preparation process. Images of blood cells are taken from the prepared slide using a microscope of Nikon Eclipse C*i* and a digital color camera of Nikon DS-Ri2. The microscope and camera are coupled and this arrangement is linked to a computer. Each image is captured in RGB (red–green–blue) color space, which is a three-channel color format. The color depth in images is 8 bits for each color channel. Images are captured at, 400× and 1000× magnifications, which correspond to objective lenses 40×, and 100×, respectively. The pathologist first identifies an ROI by observing the slide with the aid of the microscope, then snaps a digital image with a magnification of 400×. Then magnification is improved and set to 1000× which helped in obtaining a better magnified blood cells image, particularly WBC. From a single slide, on an average 19 images are taken. All captured digital images are saved without compressing with a resolution and dimension of 96 pixels per inch, and 4908 × 3264 pixels, respectively, in TIFF format.

In our study, leukemia cases of CLL, ALL, CML, and AML are considered. Subjects with other hematological/hematolymphoid malignancies are excluded from the study. Fig. 3 shows examples from the proposed dataset. Normal case samples are presented in the first row, followed by samples of CLL, ALL, CML, and AML in the second, third, fourth, and fifth rows, respectively. Images obtained at 400× magnification appear in the first column, while images obtained at 1000× magnification appear in the second and third columns.

Details of the image distribution are shown in Table 3. The database currently has 1250 images. There are 250 images of each class, *i.e.*, normal, CLL, ALL, CML, and AML. Microscopic images of two sub-types of ALL, *i.e.* T-ALL and B-ALL, are present in the dataset. The dataset also contains images of six sub-types of AML, *i.e.*, M0, M1, M2, M3, M4, and M5.

3. Deep transfer learning method for detection of leukemia and its types

Our study aims to detect the type of leukemia using microscopic blood smear images. Fig. 4(a) shows the outline of our study. The whole dataset is divided into two unequal parts, training dataset and testing dataset. The frozen convolutional layers, also known as convolutional bases, of pre-trained CNN models are used for feature extraction from microscopic blood smear images. EFs are used to train a classifier to classify the label of a given input image. This technique to train a classifier is called deep transfer learning [49]. This technique is very useful when the dataset for training a CNN model is not large, very common in the case of datasets belonging to the biomedical field.

In our study, SVM [32], RF [30], and a new set of FCL on the top of the convolutional base of different pre-trained CNN models are used as classifiers. Frozen convolutional bases of MobileNet [50], DenseNet121 [51], ResNet152V2 [52], VGG16 [53], Xception [54], and InceptionV3 [55] are used as feature extractors. These CNN models are already trained on huge datasets like the ImageNet dataset to the highest possible accuracy, are publicly available, and are easily accessible [49].

During the training of a CNN model, the kernels of convolutional layers obtain the most optimized values with the help of an optimization algorithm corresponding to a specific dataset. The earlier convolutional layers extract basic and highly generic features like edges, textures, and colors in an image. These features are generally local and not specific to a dataset. As the depth of the layers goes on increasing they start extracting features that are more specific and related to the training dataset [49]. Thus, when a new dataset, which is not related to the training dataset, is used for classification task using deep transfer learning then it will be much more beneficial to not use the few last convolutional layers of the pre-trained CNN models for feature extraction. The kernels of the few last convolutional layers need to be updated with the most optimized values corresponding to the new dataset. This process of updating a few last convolutional layers of pre-trained CNN models is called fine-tuning of pre-trained CNN models. Hence, in our study, to further enhance the classification capability of FCL, kernels of a few last convolutional layers of pre-trained CNN models are also fine tuned along with FCL. Fig. 4(b) shows the fine tuning of LTCL plus FCL of VGG16 [53]. VGG16 consists of 13 convolutional layers followed by 3 FCL (including the output layer). The first two convolutional blocks consist of two convolutional layers followed by a max-pooling layer and the last three convolutional blocks consist of three convolutional layers followed by a max-pooling layer. The pooling layer helps in reducing the size of the feature maps and thus, helps in reducing the number of trainable parameters. In the output layer, softmax activation is utilized to predict the output label.

As LTCL plus FCL of VGG16 is fine-tuned on the proposed leukemia image dataset, the features extracted by LTCL will be more specific to the leukemia dataset rather than the ImageNet dataset. Training a classifier on these features will help in enhancing the performance of the classifier for the classification of leukemia. Fig. 4(c) shows

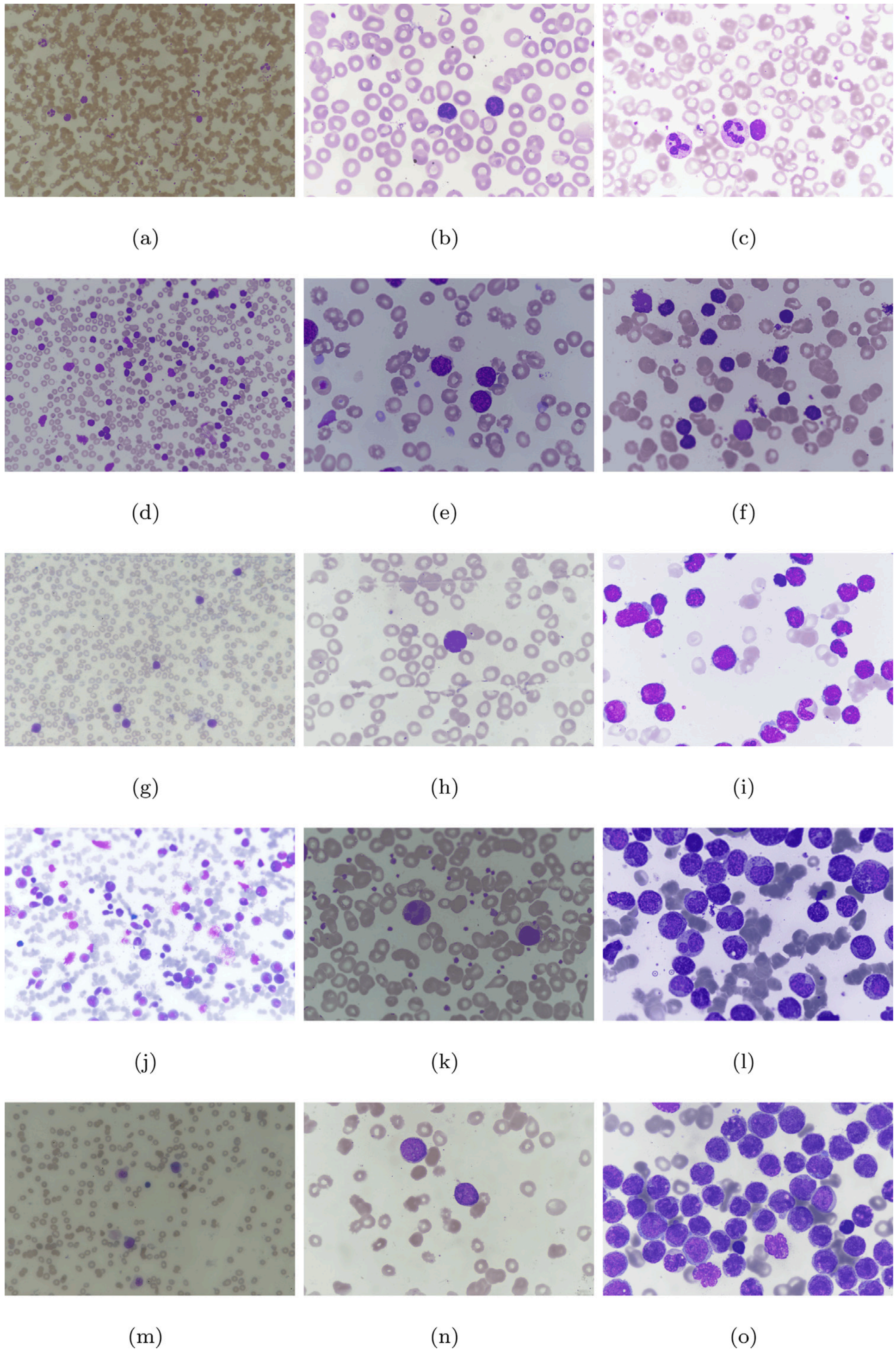


Fig. 3. Sample images of the proposed combined dataset. (a)–(c) normal cases (d)–(f) CLL (g)–(i) ALL (j)–(l) CML (m)–(o) AML. Images of 400× magnification are in the first column, while images of 1000× magnification are in the second and third columns.

convolutional bases of previous fine tuned VGG16 (LTCL plus FCL) are used for feature extraction. EFs are concatenated and used for the training and testing of SVM and RF. This method helped in enhancing the performance of SVM and RF which is shown in the ensuing section.

SVM [32] as a classifier can handle both regression and classification problems. This classifier uses a hyperplane to divide data points into different classes. The hyperplane gets optimized, based on features of data points, during training of the SVM.

RF [30] as a classifier can also perform classification and regression tasks. It takes decisions by aggregating the output of numerous decision trees.

4. Experimental results and discussions

In our study, the combined dataset is used to train the classifiers and evaluate their performance. The trained classifiers are evaluated

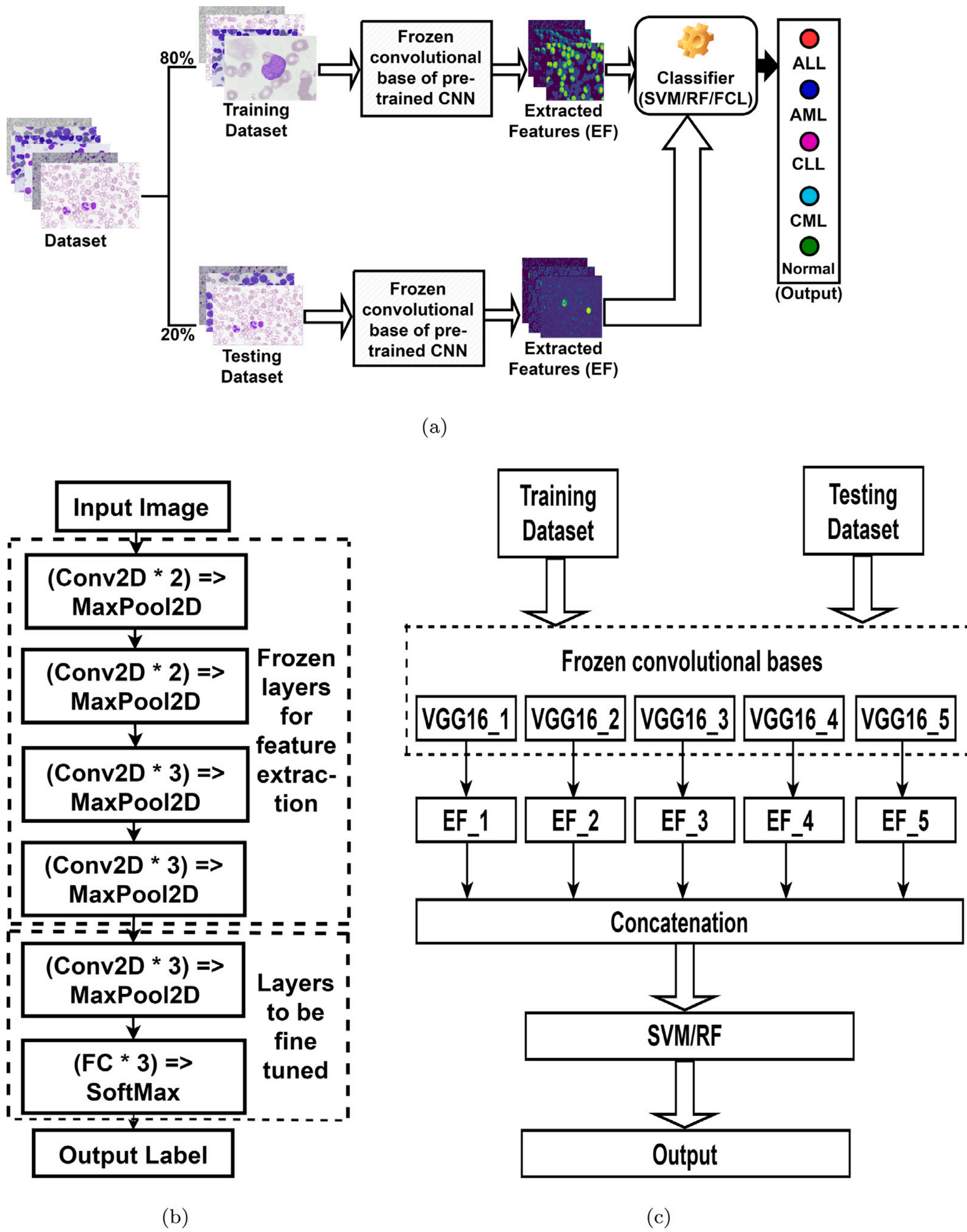


Fig. 4. (a) Outline of the proposed methodology, (b) fine tuning of LTCL plus FCL of VGG16, and (c) concatenation of extracted features to train and test SVM and RF.

on subject-independent and subject-dependent test datasets. Subject-independent test dataset contains images that do not belong to those subjects whose images are present in the training dataset. This helps in ensuring that the classifiers function well against unseen data. Subject-dependent test dataset is formed by randomly shuffling and splitting the whole dataset into two parts, *i.e.*, the training dataset, and the test dataset. There is a high chance that the images present in the subject-dependent test dataset may belong to those subjects whose images are also present in the training dataset.

The combined dataset is divided into training and test datasets in two different ways, *i.e.*,

1. 80% data is used for training and 20% data is used for testing.
2. 50% data is used for training and 50% data is used for testing.

Table 4 shows data splitting details for SVM, RF, and FCL used as classifiers when 80% data is used for training and 20% data is used for testing. For FCL, the validation dataset images, *i.e.* 25 images corresponding to each class, are randomly selected from the training dataset.

Table 4
Data splitting details when 80% data is used for training and 20% data is used for testing.

Class	SVM and RF		FCL		
	Train	Test	Train	Validation	Test
ALL	200	50	175	25	50
AML	200	50	175	25	50
CLL	200	50	175	25	50
CML	200	50	175	25	50
Normal	200	50	175	25	50
Total	1000	250	875	125	250

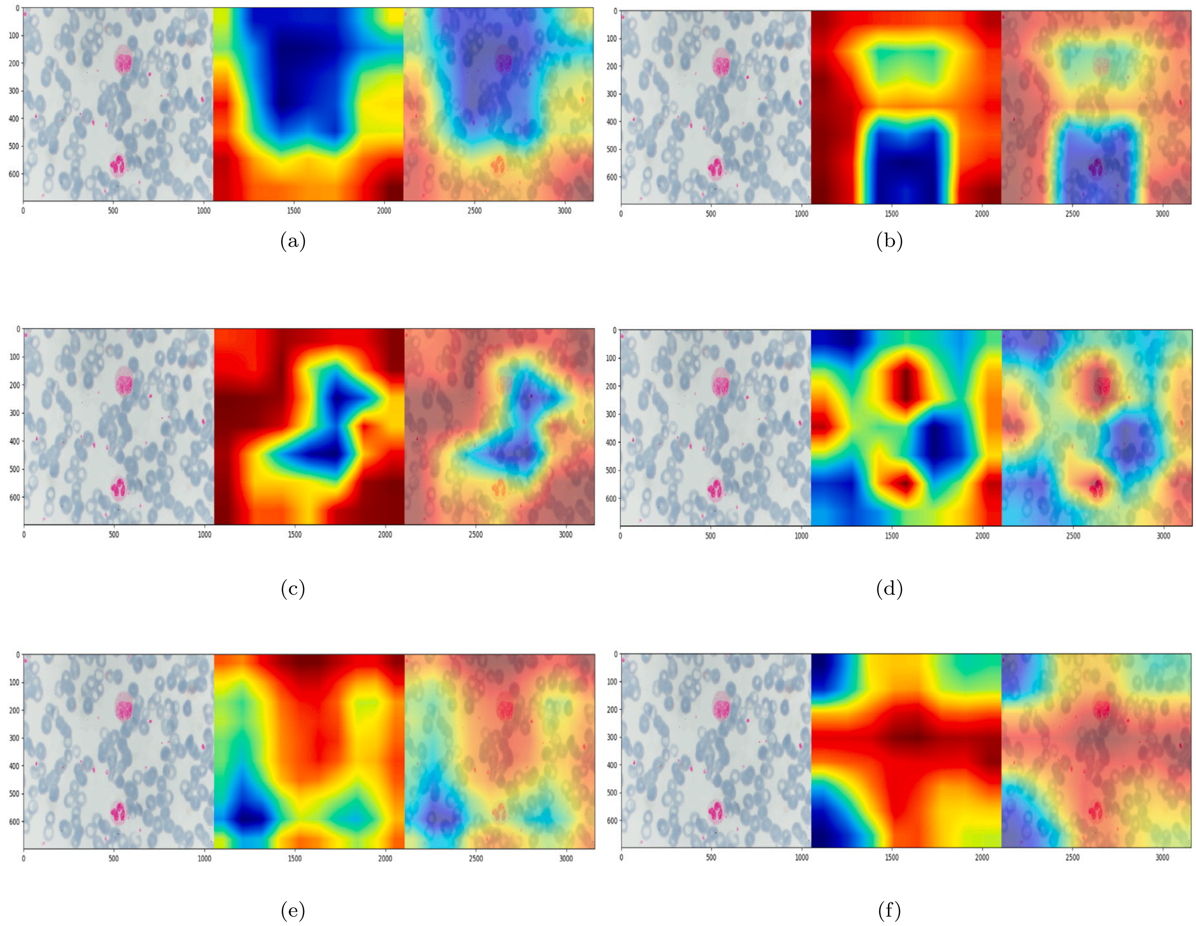


Fig. 5. Grad-CAM results obtained for (a) MobileNet (b) DenseNet121 (c) ResNet152V2 (d) VGG16 (e) Xception and (f) InceptionV3. The first image (left) is the original input image, the middle image is the class activation heatmap, and the third image (right) is the class activation heatmap superimposed over the original image.

Table 5
Data splitting details when 50% data is used for training and 50% data is used for testing.

Class	SVM and RF		FCL		
	Train	Test	Train	Validation	Test
ALL	125	125	115	10	125
AML	125	125	115	10	125
CLL	125	125	115	10	125
CML	125	125	115	10	125
Normal	125	125	115	10	125
Total	625	625	575	50	625

Table 5 shows data splitting details for SVM, RF, and FCL used as classifiers when 50% data is used for training and 50% data is used for testing. For FCL, the validation dataset images, *i.e.* 10 images corresponding to each class, are randomly selected from the training dataset.

The images were resized to 224×224 for MobileNet [50], DenseNet121 [51], ResNet152V2 [52], VGG16 [53], and to 299×299 for Xception [54] and InceptionV3 [55].

For RF and SVM, a 5-fold cross-validation method was used in the study. Five trials were taken for FCL and the final test accuracy of the trained models is obtained by taking the average of the five test results. The grid search method was employed to determine the type of kernel and the value of hyperparameter C of SVM. Linear kernel and $C = 0.1$ produced the best results corresponding to SVM as a classifier. The number of decision trees, *i.e.* n, in the case of RF was also determined by employing the grid search method. All the results corresponding to SVM and RF were obtained with the help of scikit-learn [56].

The images in the combined dataset are acquired using a microscope. The focus of the microscope is adjusted manually, hence the quality of each image is not the same. Therefore, there is a need to normalize images so that fine tuning of CNN models takes place properly [57,58]. To normalize images, the mean subtraction method is utilized. Mean values corresponding to each channel are calculated and subtracted from every pixel in the channel. Then images are resized as per the input image size specification of the CNN models used. The FCL added on the top of the convolutional base of CNN models has an output layer of size five. Softmax activation is utilized in the final layer. For training, various epochs, batch sizes, initial learning rates, and optimizers were explored; the best result came from 20 epochs, batch size of 10, 0.0001 initial learning rate, and stochastic gradient descent [59] optimizer with 0.9 momentum. To avoid overfitting, augmentation of the training dataset was done, and the learning rate was linearly reduced. Trained models were evaluated using the test datasets and corresponding test accuracies were obtained. The studies were carried out with an Nvidia RTX 2080 Ti graphics processing unit having 11 GB memory, and the Keras framework [60].

Tables 6–8 show the performance of RF, SVM, and FCL along with different state-of-the-art pre-trained CNN models for different data splits. Test accuracies corresponding to subject-independent 80–20 data split for RF, SVM, and FCL along with pre-trained VGG16 are 72%, 81%, and 74%, respectively. The number of decision trees, *i.e.* n, is 600.

Accuracy is given by

$$\text{Accuracy} = \frac{\text{Total number of samples correctly predicted}}{\text{Total number of samples in the test dataset}} \tag{1}$$

Table 6
Test accuracies (%) corresponding to different CNN models along with RF for different data splits.

S.No.	Model	Subject-independent		Subject-dependent	
		80 – 20	50 – 50	80 – 20	50 – 50
1	MobileNet	53 (n = 300)	38 (n = 100)	79 (n = 800)	76 (n = 800)
2	DenseNet121	56 (n = 800)	37 (n = 400)	83 (n = 700)	83 (n = 500)
3	ResNet152V2	50 (n = 700)	33 (n = 100)	72 (n = 200)	70 (n = 200)
4	VGG16	72 (n = 600)	47 (n = 300)	83 (n = 400)	78 (n = 400)
5	Xception	52 (n = 600)	29 (n = 600)	68 (n = 900)	66 (n = 400)
6	InceptionV3	42 (n = 500)	27 (n = 600)	67 (n = 200)	64 (n = 200)

Table 7
Test accuracies (%) corresponding to different CNN models along with SVM for different data splits.

S.No.	Model	Subject-independent		Subject-dependent	
		80 – 20	50 – 50	80 – 20	50 – 50
1	MobileNet	57	40	86	83
2	DenseNet121	72	43	86	84
3	ResNet152V2	64	36	75	73
4	VGG16	81	55	88	85
5	Xception	68	39	72	69
6	InceptionV3	52	33	75	69

Table 8
Test accuracies (%) corresponding to different CNN models along with FCL for different data splits.

S.No.	Model	Subject-independent		Subject-dependent	
		80 – 20	50 – 50	80 – 20	50 – 50
1	MobileNet	60	32	68	65
2	DenseNet121	61	43	81	72
3	ResNet152V2	54	35	62	56
4	VGG16	74	51	83	84
5	Xception	51	43	74	75
6	InceptionV3	43	32	72	72

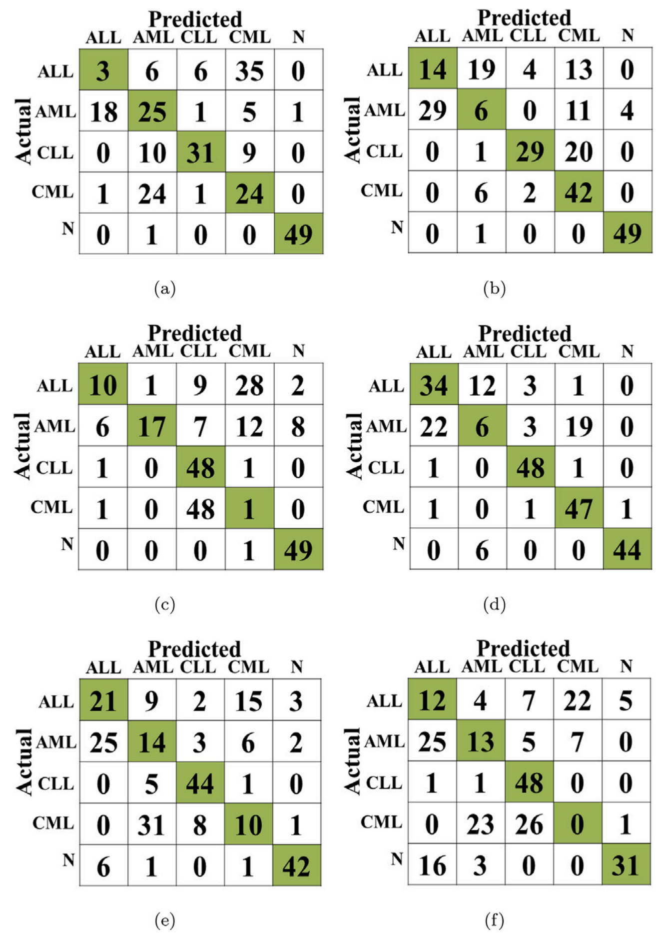


Fig. 6. Confusion matrices obtained for RF along with (a) MobileNet (b) DenseNet121 (c) ResNet152V2 (d) VGG16 (e) Xception and (f) InceptionV3. N represents normal cases.

Test accuracies corresponding to subject-dependent 80−20 data split for RF, SVM, and FCL along with pre-trained VGG16 are 83%, 88%, and 83%, respectively, and for subject-dependent 50−50 data split are 78%, 85%, and 84%, respectively. However, test accuracies corresponding to subject-independent 50 − 50 data split for RF, SVM, and FCL along with pre-trained VGG16 reduce to 47%, 55%, and 51%, respectively. Thus, it can be inferred that the classifiers do not get properly trained using a 50−50 data split. Although performance of the classifiers corresponding to subject-dependent 50 − 50 data split is better than that of subject-independent 50 − 50 data split but the trained models cannot perform well against a new dataset taken from an unknown subject. Hence, a model should be trained on the subject-independent dataset to make it robust against a new and unseen dataset.

The performance of the classifiers against subject-independent 80 − 20 test data is the best when they are trained on features extracted by the convolutional base of VGG16. Fig. 5 shows the heatmaps obtained by Grad-CAM [61] technique. Grad-CAM is a gradient-based localization technique that is used to visually inspect which portion or pattern of an input image is given more weightage by a CNN model while classifying the image to a particular class c. The weightage given to a particular feature map A^K for class c is given by

$$w_K^c = \frac{1}{\mathrm{N}} \sum_p \sum_q \frac{\partial Y^c}{\partial A_{pq}^K} \tag{2}$$

where $\frac{\partial Y^c}{\partial A_{pq}^K}$ is gradient of the score Y^c with respect to feature map A^K of a convolutional layer and the rest portion in the right hand side of Eq. (2) represents global-average-pooling action done over width and height dimensions of a feature map (indexed by p and q respectively). N is the number of pixels in the feature map.

Then Grad-CAM for a class c is obtained by a weighted combination of the feature maps,

$$\phi_{\text{Grad-CAM}}^c = \mathrm{ReLU}\left(\sum_K w_K^c A^K\right) \tag{3}$$

ReLU is applied to focus on the pixels which belong to the class of interest.

This technique can be used for CNN model interpretability and qualitative analysis of obtained results by the models. In Fig. 5, the first image (left) is the original input image, the middle image is the class activation heatmap, and the third image (right) is the class activation heatmap superimposed over the original image. For the heatmap, a jet color scheme is used in which dark red and red color tones represent higher weightage followed by yellow, green, and blue color tones. The blue color tone represents the lowest weightage value. Fig. 5(d) shows the Grad-CAM activation map corresponding to VGG16. From the Grad-CAM activation map, it can be observed that the regions near the two

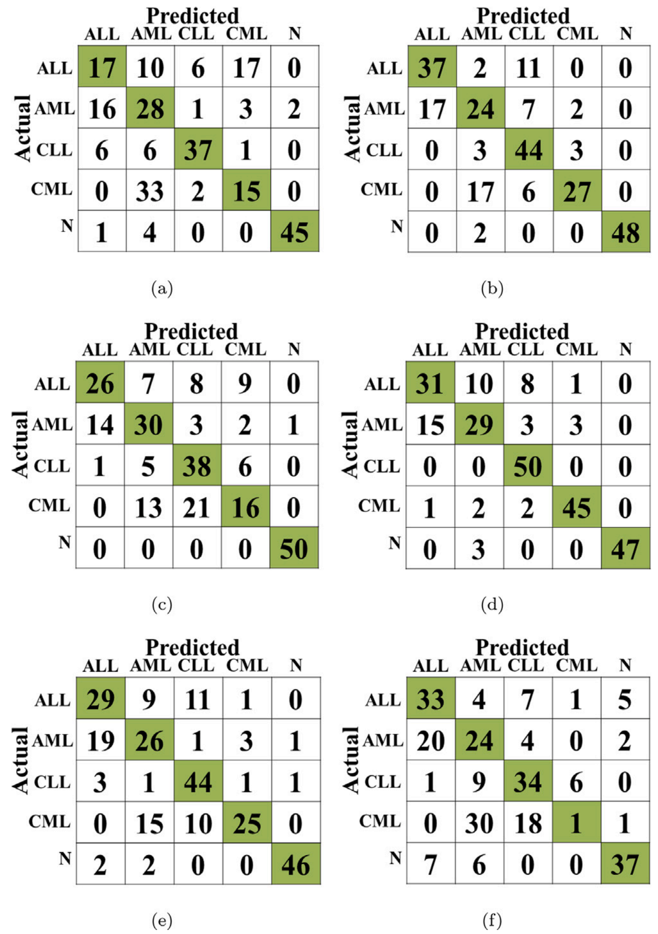


Fig. 7. Confusion matrices obtained for SVM along with (a) MobileNet (b) DenseNet121 (c) ResNet152V2 (d) VGG16 (e) Xception and (f) InceptionV3. N represents normal cases.

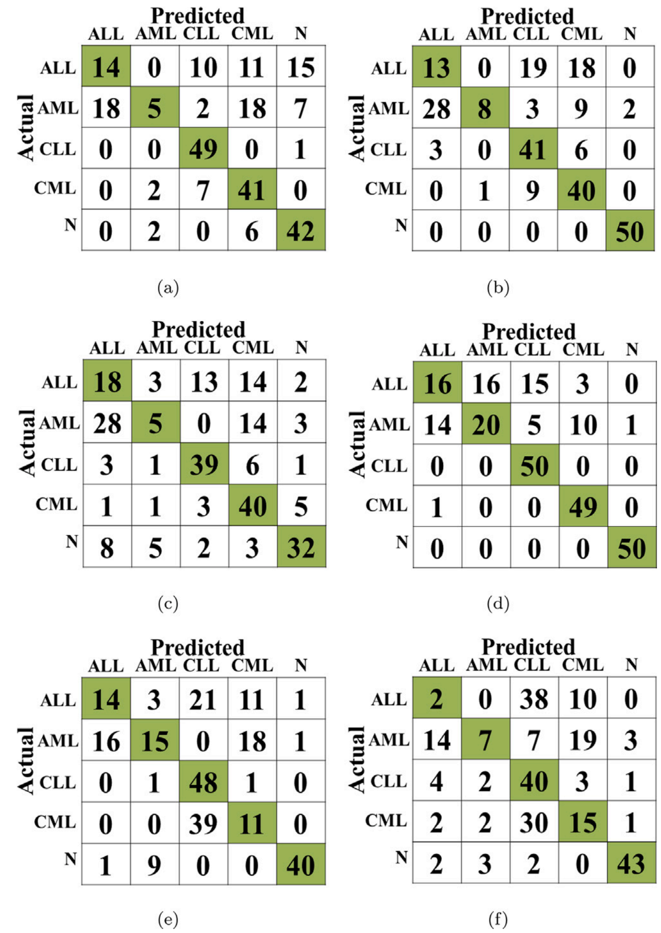


Fig. 8. Confusion matrices obtained for FCL along with (a) MobileNet (b) DenseNet121 (c) ResNet152V2 (d) VGG16 (e) Xception and (f) InceptionV3. N represents normal cases.

WBCs, *i.e.* ROI, are highly activated in comparison to other regions in the image. However, the activation maps corresponding to other CNN models are either less activated near ROI or equally activated in other regions in addition to ROI. This results in a low accuracy value while predicting the class of the input image.

Figs. 6, 7 and 8 represent confusion matrices corresponding to subject-independent $80-20$ test data when state-of-the-art pre-trained CNN models are used as feature extractors and RF, SVM, and FCL are used as classifiers, respectively. The principal diagonal elements of each confusion matrix represent the number of samples correctly predicted corresponding to a class. It is evident from the confusion matrices that the classifiers perform better when they are trained on features extracted by VGG16. To further analyze the performance of the classifiers along with VGG16, accuracy, precision, recall, specificity, and F1-score were computed using elements of the confusion matrices.

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}} \tag{4}$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{5}$$

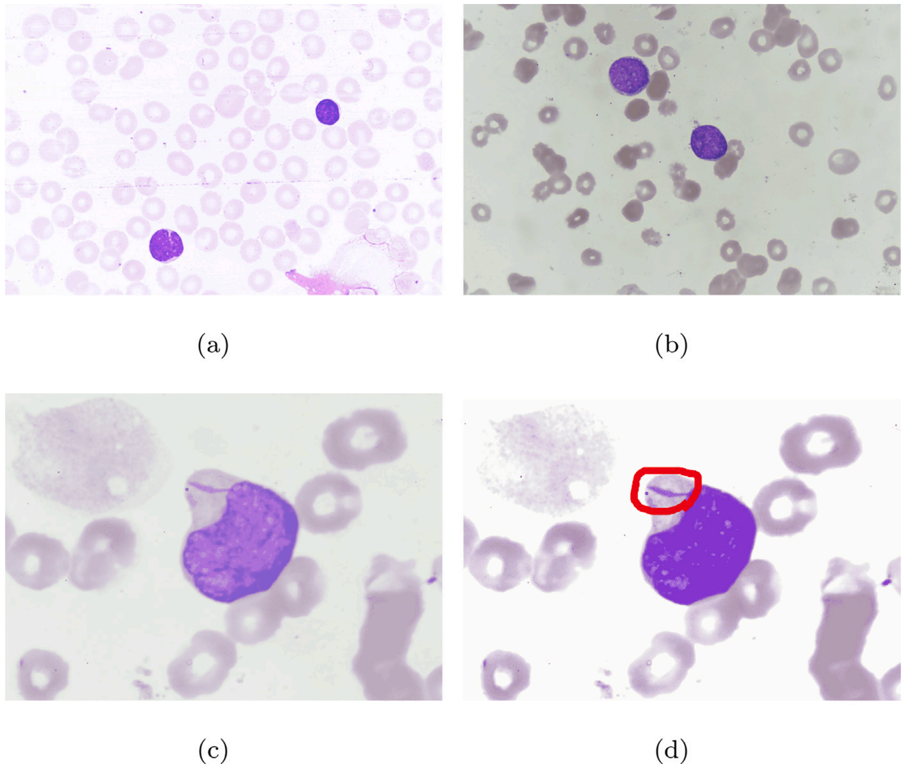
$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{6}$$

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} \tag{7}$$

$$\text{F1 score} = \frac{2\times \text{True Positive}}{(2\times \text{True Positive}) + \text{False Positive} + \text{False Negative}} \tag{8}$$

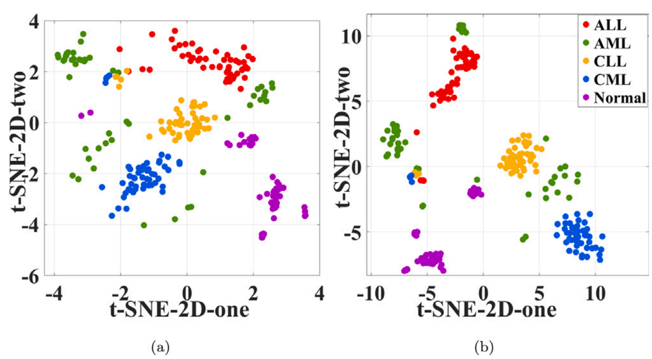
True positive for a particular class represents the number of samples classified to that class which actually belong to that class. False positive for a particular class represents the total number of samples misclassified to that particular class. False negative for a particular class represents the total number of samples of that particular class that are misclassified to other classes. True negative for a particular class is calculated by subtracting the sum of true positive, false positive, and false negative from the total number of samples present in the test dataset.

Table 9, Table 10, and Table 11 represent test values of different classification metrics obtained for RF, SVM, and FCL, respectively. It can be observed that the test values of classification metrics corresponding to CLL, CML, and normal cases are better compared to ALL and AML cases for all three classifiers. Thus, it can be inferred that CLL, CML, and normal cases are well predicted compared to ALL and AML. This is because some of the abnormal WBCs (blasts) of AML especially those belonging to the M0 sub-type of AML morphologically resemble the lymphoblasts of ALL. In most of the cases, the blasts belonging to AML are marginally larger compared to ALL. The AML blasts have relatively more abundant cytoplasm, finer nuclear chromatin, and a greater number of nucleoli compared to lymphoid blasts. Most of the sub-type of AML blasts have a very distinctive cytoplasmic feature called Auer rods, as highlighted in red in Fig. 9(d). But M0 sub-type of AML has less conspicuous nucleoli, scanty cytoplasm, and does not have



(a) (b) (c) (d)

Fig. 9. Sample belonging to (a) ALL (b) M0 AML (c) AML containing Auer rod and (d) Auer rod highlighted in red (manually done).



(a) (b)

Fig. 10. The t-SNE maps when (a) VGG16 and (b) VGG16 with LTCL fine tuned, are used as feature extractors.

Auer rods in their cytoplasm [42,62]. By definition, AML-M0 contains acute leukemias that do not meet the morphological and cytochemical criteria for AML diagnosis but can be assigned to a myeloid lineage using an immunological assay, and for which myeloid lineage assignment can be made by immunological inspection [63]. Fig. 9(a) shows a sample of ALL while Fig. 9(b) shows a sample of M0 sub-type of AML. As can be seen from Figs. 9(a) and 9(b) both look morphologically very similar to each other. So, the performance of the classifiers degrades while classifying ALL and AML cases. Fig. 9(c) shows a sub-type of AML blast, other than M0, containing an Auer rod.

Fig. 10(a) shows t-SNE [64] plot when VGG16 is used to extract features. The t-SNE is a qualitative indicator that aids in the visualization of high-dimensional data, such as images, in two or three-dimensional maps. Fig. 10(a) shows that some of the data points associated with AML and ALL are difficult to classify compared to CLL, CML, and normal cases.

When the kernels corresponding to LTCL of VGG16 are fine tuned, then a classification accuracy of 80% is achieved by FCL corresponding to subject-independent $80-20$ test data. The convolutional bases of these VGG16 models are used for feature extraction of training and test datasets. The EFs are concatenated and used for the training and evaluation of RF and SVM classifiers. Accuracies of 76% and 84% are achieved corresponding to RF and SVM, respectively. The number of decision trees, *i.e.* n, in the case of RF is 600. For subject-dependent $80-20$ test data, classification accuracies of 85%, 88%, and 88%

Table 9
Classification metrics obtained for pre-trained VGG16 along with RF.

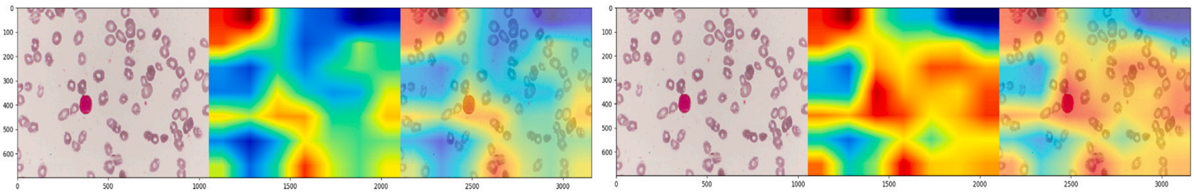
Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	84.0	0.59	0.68	0.88	0.63
AML	75.2	0.25	0.12	0.91	0.16
CLL	96.4	0.87	0.96	0.97	0.91
CML	90.4	0.69	0.94	0.89	0.80
Normal	97.2	0.98	0.88	0.99	0.93

Table 10
Classification metrics obtained for pre-trained VGG16 along with SVM.

Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	86.0	0.66	0.62	0.92	0.64
AML	85.6	0.66	0.58	0.93	0.62
CLL	94.8	0.79	1.00	0.93	0.88
CML	96.4	0.92	0.90	0.98	0.91
Normal	98.8	1.00	0.94	1.00	0.97

Table 11
Classification metrics obtained for pre-trained VGG16 along with FCL.

Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	80.4	0.52	0.32	0.93	0.40
AML	81.6	0.56	0.40	0.92	0.47
CLL	92.0	0.71	1.00	0.90	0.83
CML	94.4	0.79	0.98	0.93	0.87
Normal	99.6	0.98	1.00	0.99	0.99



(a) (b)

Fig. 11. Grad-CAM results obtained for (a) VGG16 and (b) VGG16 with LTCL fine tuned. The region near WBCs is less activated in the case of VGG16 than VGG16 with LTCL fine tuned. This results in a difference in the accuracy of the predicted class. For example, the accuracy while predicting the class of the input image, *i.e.* ALL, by VGG16 is only 55.3% while by VGG16 with LTCL fine tuned is 96.6%.

(a)

Actual \ Predicted	ALL	AML	CLL	CML	N
ALL	38	8	4	0	0
AML	24	10	2	14	0
CLL	0	0	50	0	0
CML	1	0	2	47	0
N	0	4	0	1	45

(b)

Actual \ Predicted	ALL	AML	CLL	CML	N
ALL	36	5	8	1	0
AML	13	30	3	4	0
CLL	0	0	50	0	0
CML	1	3	1	45	0
N	0	2	0	0	48

(c)

Actual \ Predicted	ALL	AML	CLL	CML	N
ALL	25	16	8	1	0
AML	16	26	5	3	0
CLL	0	0	50	0	0
CML	0	2	0	48	0
N	0	0	0	0	50

Fig. 12. Confusion matrices created with (a) RF (b) SVM and (c) FCL as classifiers, when LTCL of VGG16 are fine tuned. N represents normal cases.

Table 12
Classification metrics corresponding to RF and LTCL of VGG16 fine tuned.

Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	85.2	0.60	0.76	0.87	0.67
AML	79.2	0.45	0.20	0.94	0.28
CLL	96.8	0.86	1.00	0.96	0.93
CML	92.8	0.70	0.94	0.93	0.84
Normal	98.0	1.00	0.90	1.00	0.95

Table 13
Classification metrics corresponding to SVM and LTCL of VGG16 fine tuned.

Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	88.8	0.72	0.72	0.93	0.72
AML	88.0	0.75	0.60	0.95	0.67
CLL	95.2	0.81	1.00	0.94	0.89
CML	96.0	0.90	0.90	0.97	0.90
Normal	99.2	1.00	0.96	1.00	0.98

are achieved by RF (n = 400), SVM, and FCL, respectively. The improvement in the performance of the classifiers can be inferred from Fig. 10(b) which shows a t-SNE map when VGG16 with LTCL fine tuned is used as a feature extractor. The data points associated with different classes are more clearly separated than the corresponding data points in Fig. 10(a).

Fig. 11 shows the Grad-CAM activation map corresponding to VGG16 and VGG16 with LTCL fine tuned. From the Grad-CAM activation maps, it can be observed that the region near ROI (WBCs) is

Table 14
Classification metrics corresponding to FCL and LTCL of VGG16 fine tuned.

Class	Accuracy(%)	Precision	Recall	Specificity	F1 Score
ALL	83.6	0.61	0.50	0.92	0.55
AML	83.2	0.59	0.52	0.91	0.55
CLL	94.8	0.79	1.00	0.93	0.88
CML	97.6	0.92	0.96	0.98	0.94
Normal	100.0	1.00	1.00	1.00	1.00

less activated in the case of VGG16 than VGG16 with LTCL fine tuned. This results in a difference in the accuracy of the predicted class. For example, the accuracy while predicting the class of the input image, *i.e.* ALL, shown in Fig. 11(a) is only 55.3% while for Fig. 11(b) is 96.6%.

Fig. 12 shows the confusion matrices for RF, SVM, and FCL when the kernels corresponding to LTCL of VGG16 are fine tuned. From the confusion matrices, it can be inferred that the number of correctly predicted ALL and AML samples increased along with CLL, CML, and normal samples. Fig. 13 shows performance comparison in terms of test accuracy for RF, SVM, and FCL when features are extracted by convolutional bases of pre-trained VGG16, and VGG16 with LTCL fine tuned. It can be observed that the performance of the classifiers improves when they are trained on features extracted by VGG16 with LTCL fine tuned. Fig. 13(d) shows performance comparison in terms of overall test accuracy for all three classifiers when features are extracted by convolutional bases of pre-trained VGG16, and VGG166 with LTCL fine tuned. The overall test accuracies of the classifiers also improved corresponding to VGG16 with LTCL fine tuned.

Table 12, Table 13, and Table 14 represent different classification metrics obtained for RF, SVM, and FCL, respectively. The test values of classification metrics also improved for ALL and AML cases along with CLL, CML, and normal cases. From Table 13 it can be observed that the test accuracies obtained by the proposed method corresponding to ALL and AML cases are 88.8% and 88.0%, respectively. From Table 14 it can be observed that the test values corresponding to the normal case reached 100%. For CLL and CML cases, test accuracies reached 94.8% and 97.6%, respectively.

Fig. 14 shows Grad-CAM results obtained for samples belonging to normal, CLL, ALL, CML, and AML classes. The samples are classified using VGG16 with LTCL fine tuned. From the Grad-CAM activation maps, it can be observed that the regions near WBCs and other important features which help in detecting the type of leukemia are more activated. For example, in Fig. 14(b) SCs (encircled in blue) along with WBCs are also covered by the heatmap. SCs have important biologic correlations with CLL cases [65]. Similarly, in Fig. 14(d) platelets (marked by black arrows) and WBCs are covered by the corresponding heatmap. It is seen that patients with CML have too many platelets (thrombocytosis) [66].

Table 15 shows a comparison of the works present in the literature and the proposed work for the classification of leukemia. Ahmed et al. [67] have achieved an accuracy of 81.74% in identifying normal cases and the four major types of leukemia, *i.e.*, CLL, ALL, CML, and AML. They trained a CNN model and evaluated its performance using

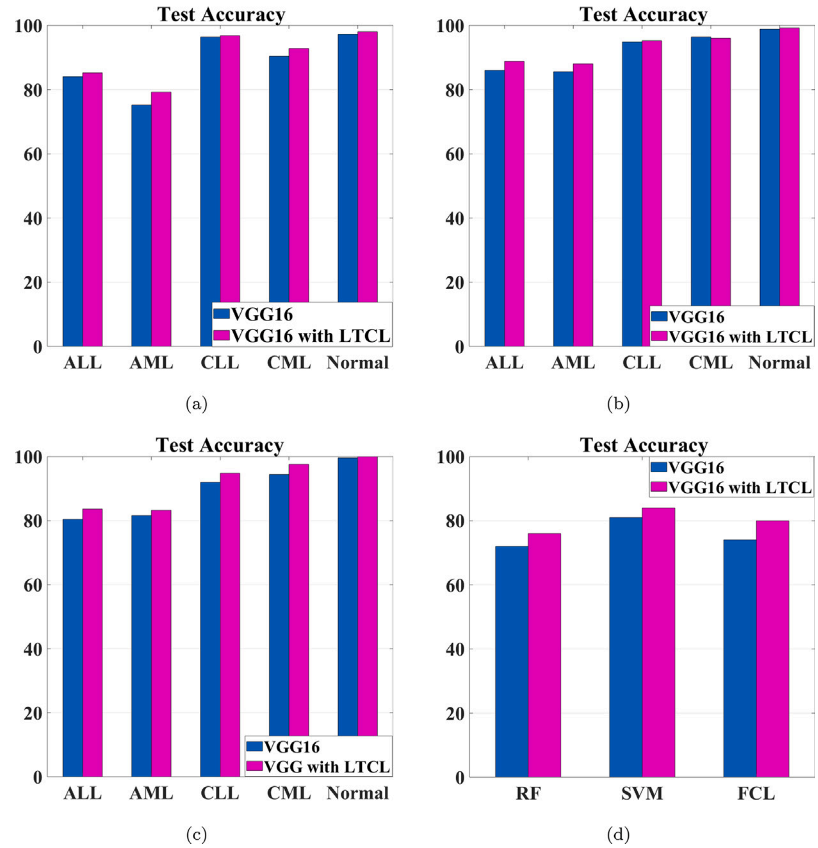


Fig. 13. Performance comparison for (a) RF (b) SVM (c) FCL, and (d) all the three classifiers, when features are extracted by convolutional bases of pre-trained VGG16, and VGG16 with LTCL fine tuned.

Table 15
Comparison of the works present in the literature and the proposed work for classification of leukemia.

Authors	Classifier	Dataset size	Type	Accuracy(%)
Ahmed et al. [67] (2019)	CNN	903	CLL, ALL, CML, AML, and normal	81.74
Bibi et al. [68] (2020)	ResNet-34	518	CLL, ALL, CML, AML, and normal	99.56
	DenseNet-121			99.91
Dese et al. [20] (2021)	Multi class SVM	520	CLL, ALL, CML, AML, and normal	97.69
Baig et al. [26] (2022)	Concatenated features along with bagging ensemble	452	MM, ALL, and AML	97.04
Proposed work	**SVM (VGG16 along with SVM)**	**1250**	**CLL, ALL, CML, AML, and normal**	**81.00**
	SVM (VGG16 with LTCL fine tuned along with SVM)			**84.00**

a dataset containing only 903 images. Their work also does not mention the evaluation of the trained CNN model on a subject-independent test dataset, CNN model interpretability, and qualitative analysis of obtained results. Bibi et al. [68] have obtained classification accuracies of 99.56% and 99.91% using an IoMT-based ResNet-34 and DenseNet-121 for leukemia classification. They used a dataset having only 518 images and to overcome the problem of a small dataset they applied data augmentation to the whole dataset. The augmented dataset was split into training and test datasets. Hence, the trained models were evaluated on the augmented test dataset. Consequently, the obtained classification accuracies on augmented test dataset images cannot be realistic [24]. Dese et al. [20] have achieved an accuracy of 97.69% in detecting normal, CLL, ALL, CML, and AML cases. They used a dataset containing 520 microscopic blood smear images. Pre-processing of images is done to improve image quality and segmentation is done to remove unwanted objects present in the images. From segmented images, texture, geometric, statistical, and color features are extracted and used to train a multi class SVM to classify and detect leukemia. Baig et al. [26] employ transfer learning technique to detect MM, AML, and ALL and have achieved 97.04% accuracy. They used a dataset having only 452 images out of which 114 images belong to MM, 31 images belong to ALL, and 307 images belong to AML. To overcome the problem of a small number of images belonging to MM and ALL they applied data augmentation. The dataset containing images of AML along with augmented images of MM and ALL was split into training and test datasets. Hence, the trained model, *i.e.* bagging ensemble, was evaluated on the test dataset containing augmented images of MM and ALL cases. Similar to Bibi et al. [68], the obtained classification accuracy on augmented test dataset images cannot be realistic. From Table 15, it can be analyzed that the proposed method using SVM along with pre-trained VGG16 obtained an accuracy of 81%. The accuracy got better and improved to 84% when a combination of SVM and VGG16

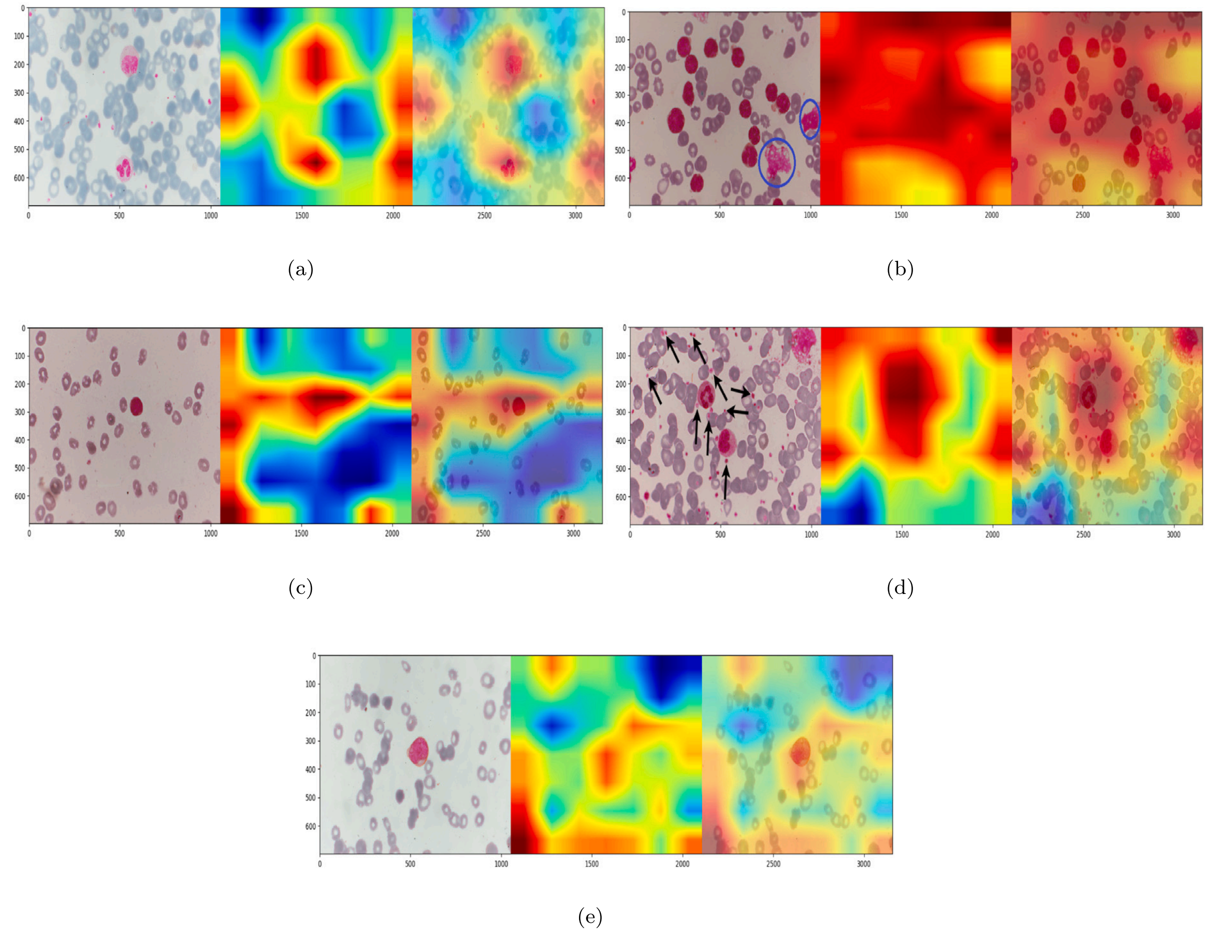


Fig. 14. Grad-CAM results obtained for samples belonging to (a) Normal (b) CLL (smudge cells are encircled in blue) (c) ALL (d) CML (some of the platelets are shown by black arrows) and (e) AML.

with LTCL fine tuned is used for the classification of normal cases and four types of leukemia on a subject-independent dataset. Thus, the proposed work achieved the second-best performance after Dese et al. [20] regarding the classification of normal and four major types of leukemia. However, in the proposed work no segmentation technique is applied to the images as done in Dese et al. [20]. In the future, a suitable segmentation technique can be applied to the microscopic images to remove unwanted regions which may help in achieving better accuracy in detecting leukemia and its types. The results obtained in the proposed work are also analyzed qualitatively with the help of t-SNE plot and heatmaps.

4.1. Discussion

Recently, deep transfer learning has been applied in many medical image analysis studies, especially those based on deep learning techniques. Pre-trained CNN models can be fine-tuned easily and faster, even on a distinct dataset, than training an untrained CNN model with randomly initialized weights from scratch [69]. Deniz et al. [70] classified the histopathological breast cancer images of BreaKHis database [71] into benign and malignant classes by fine tuning pre-trained AlexNet [72]. They achieved an accuracy of 91.37% for the classification task. Ohata et al. [73] applied deep transfer learning technique to detect and classify eight different categories of colorectal cancer using a histology dataset that contains 5000 images. The features extracted by pre-trained DenseNet169 were used to train an SVM for the classification task and an accuracy of 92.083% was achieved. Ferreira et al. [74] classified histological images of breast cancer into four different classes, *i.e.* normal tissue, *in situ* carcinoma, benign tumor, and invasive carcinoma, by fine tuning few feature extraction layers of Inception ResnetV2. The fine tuned model was evaluated on a test dataset containing 100 images and accuracy of 76% was achieved. Khobragade et al. [75] used deep transfer learning for automated screening of cervical cancer cells. Deep EfficientNet model initially trained on the ImageNet dataset was used to detect cervical cancer cells on a dataset of 500 images and obtained an accuracy of 98.40%. George et al. [76] proposed a nucleus guided transfer learning technique to detect breast cancer using BreaKHis database [71]. The proposed framework obtained 96.91% accuracy to classify benign and malignant samples.

In our study, deep transfer learning has also been used for the detection and classification of leukemia. However, deep transfer learning technique suffers from negative transfer drawback due to the difference between the domain of the training dataset used for training the CNN models and the dataset used during deep transfer learning [77]. Since the pre-trained CNN models used in our study are trained on the ImageNet dataset that contains only real-life images, so there is a chance of negative transfer as a microscopic blood smear dataset is used during deep transfer learning in our study. In the future, the availability of a large microscopic blood smear dataset can help in training a deep learning model from scratch and achieving higher accuracy in detecting leukemia and its types.

5. Conclusion

In this study, the detection of leukemia and classification of its four major types, *i.e.*, CLL, ALL, AML, and CML using deep transfer learning is presented. The impact of subject-independent test data, *i.e.*

images considered in the test dataset do not belong to subjects whose images are present in the training dataset, on the robustness of trained models is also analyzed. Quantitative analysis of results achieved in this study is done with the help of obtained accuracies, precision, recall, specificity, and F1 scores of the classifiers, whereas qualitative analysis is done with the help of Grad-CAM activation map and t-SNE plot. The quantitative analysis results are consistent with their qualitative counterpart. The classification problem has accomplished an accuracy of 74% for a pre-trained VGG16 along with FCL. Pre-trained VGG16 along with SVM helped in achieving an accuracy of 81%. When LTCL of VGG16 is also fine tuned then it helped in better classification of acute type leukemia along with chronic ones. Hence, the overall classification accuracy of classifiers also improved on the combined dataset. FCL as a classifier achieved an accuracy of 80% whereas SVM as a classifier achieved an accuracy of 84%. The features responsible for classifying an image to a particular class are visualized with the help of class-specific heatmaps generated by Grad-CAM technique. Availableness of the proposed combined dataset for the scientific fraternity will assist in developing an automated diagnostic mechanism for the detection of leukemia and its types in the initial stage only. Such a mechanism would benefit pathologists by truncating their workload and minimizing the likelihood of errors. Future work will emphasize improving the overall classification task performance by increasing the number of labeled data of leukemia.

The source code, trained models, and the dataset are available at https://github.com/Arjuniitp/Leukemia-classification.

CRediT authorship contribution statement

Arjun Abhishek: Experimentation, Methodology, Writing – original draft. **Rajib Kumar Jha:** Supervision, Visualization, Investigation. **Ruchi Sinha:** Data collection, Evaluation. **Kamlesh Jha:** Data collection, Evaluation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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